

CUSTOMER CHURN ANALYSIS

Exploratory Data Analysis (EDA) using Python

Tools & Technologies

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Google Colab
 - Git & GitHub
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Project Type

Exploratory Data Analysis (EDA)

Dataset

IBM Telco Customer Churn Dataset

Executive Summary

Customer retention is one of the most critical challenges for subscription-based businesses, as retaining existing customers is considerably more cost-effective than acquiring new ones. Understanding the factors that influence customer churn enables organizations to improve customer satisfaction, strengthen loyalty, and increase long-term profitability.

This project performs an end-to-end Exploratory Data Analysis (EDA) on the IBM Telco Customer Churn dataset to identify the key drivers of customer churn. The analysis covers data assessment, data cleaning, feature engineering, univariate analysis, numerical analysis, and bivariate analysis to examine customer behaviour from multiple perspectives.

The findings indicate that customer churn is strongly associated with customer tenure, contract type, internet service, payment method, monthly charges, and the adoption of value-added services such as online security and technical support. Customers combining several of these high-risk characteristics were found to have a significantly greater likelihood of leaving the company.

Based on these insights, practical business recommendations are proposed to help reduce customer churn, improve customer retention, and support data-driven decision-making. These recommendations include encouraging long-term contracts, strengthening customer onboarding, promoting value-added services, and developing targeted retention strategies for high-risk customer segments.

1. Business Problem

Customer churn is a major challenge for subscription-based businesses because the loss of existing customers can negatively affect revenue, customer lifetime value, and long-term business growth.

The business needs to understand **why customers leave, who is most likely to churn, and what characteristics are associated with higher or lower churn risk.**

This project therefore focuses on analyzing customer-level data to identify the major patterns and characteristics associated with churn and translate those findings into practical actions that can help improve customer retention.

Key Business Questions

The analysis aims to answer four main questions:

1. **Why are customers churning?**
2. **Who is most likely to churn?**
3. **What characteristics are associated with higher or lower churn risk?**
4. **What actions can the business take to reduce churn and improve customer retention?**

2. Project Objectives

The main objectives of this project are to:

- Assess the structure and quality of the customer dataset.
- Clean and prepare the data for reliable analysis.
- Understand the distribution of customer characteristics.
- Analyze churn patterns across categorical variables.
- Analyze numerical variables such as tenure, monthly charges, and total charges.
- Examine relationships between multiple customer characteristics and churn.
- Identify high-risk and low-risk customer segments.
- Determine the key characteristics associated with customer churn.
- Translate analytical findings into practical customer-retention recommendations.

3. Dataset Overview

Dataset Information

Attribute	Description
Dataset	IBM Telco Customer Churn Dataset
Source	IBM Sample Dataset
Original Records	7,043 customers
Records After Cleaning	7,032 customers
Columns	21
Target Variable	Churn
Data Type	Customer-level structured data
Analysis Type	Exploratory Data Analysis

Dataset Description

The dataset contains information about customers of a telecommunications company. It includes demographic characteristics, customer relationship information, subscribed services, contract details, billing information, payment methods, and customer tenure.

The **Churn** variable indicates whether a customer has left the company and is therefore used as the primary target variable throughout the analysis.

The dataset contains both **categorical and numerical variables**, allowing customer churn to be examined from multiple perspectives.

Main Variable Categories

Customer Demographics

- Gender
- Senior Citizen
- Partner
- Dependents

Services & Subscriptions

- Phone Service
- Multiple Lines
- Internet Service
- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies

Account & Billing Information

- Contract
- Paperless Billing
- Payment Method
- Tenure
- Monthly Charges
- Total Charges

Target Variable

- Churn

Data Preparation Summary

The original dataset contained **7,043 customer records**. During the data-cleaning stage, **11 records with missing Total Charges values were removed**, resulting in **7,032 records** being used for the final analysis.

No duplicate records were identified, and the numerical variables were converted to appropriate numeric data types before analysis.

4. Methodology

The project followed a structured Exploratory Data Analysis (EDA) workflow to move from raw customer data to actionable business insights.

Analysis Workflow

Business Problem

↓

Data Assessment

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Data Cleaning & Preparation

↓

Feature Engineering

↓

Univariate Churn Analysis

↓

Numerical Analysis

↓

Bivariate Churn Analysis

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Key Insights & Customer Characteristics

↓

Business Recommendations

↓

Conclusion

Analysis Approach

The analysis was performed in the following stages:

1. Data Assessment

The dataset was first examined to understand its structure, dimensions, column names, data types, duplicate records, missing values, and categorical values.

2. Data Cleaning & Preparation

Data-quality issues were identified and addressed before performing any churn analysis. Numerical columns were converted to appropriate data types, missing values were handled, column names were standardized, and unnecessary identifiers were removed.

3. Feature Engineering

A tenure_bin variable was created by grouping customer tenure into six lifecycle categories:

- 0–12 months
- 13–24 months
- 25–36 months
- 37–48 months
- 49–60 months
- 61–72 months

This allowed customer churn to be examined across different stages of the customer lifecycle.

4. Univariate Churn Analysis

Categorical variables were analyzed individually against the Churn variable to understand which customer characteristics were associated with higher or lower churn.

Reusable Python iteration was used to reduce repetitive visualization code and maintain consistency across the analysis.

5. Numerical Analysis

The distributions of:

- Tenure
- Monthly Charges
- Total Charges

were examined and compared between churned and active customers. Correlation analysis was also performed to understand relationships between the numerical variables.

6. Bivariate Churn Analysis

Multiple categorical characteristics were analyzed together to identify customer segments with particularly high or low churn risk.

The analysis compared the overall customer population with churned customers and calculated churn rates for relevant customer groups.

Reusable functions were developed to automate repetitive bivariate visualizations and make the analysis more efficient and maintainable.

7. Insight Generation

The results from the different stages were compared to identify the strongest and most consistent churn patterns rather than relying on a single variable in isolation.

8. Business Recommendations

The identified churn patterns were translated into practical recommendations focused on improving customer retention and reducing customer churn.

5. Data Cleaning & Preparation

Before conducting the exploratory analysis, the dataset underwent several data-quality checks and preparation steps.

Data Quality Checks

Check	Result / Action
Duplicate Records	No duplicate records found
Column Names	Standardized to snake_case
Missing Values	11 missing Total Charges records identified
Numerical Data Types	Converted to appropriate numeric types
Categorical Values	Reviewed for consistency
Outliers	Checked using the IQR method
Customer ID	Removed before churn analysis
Feature Engineering	Created tenure_bin

Missing Value Handling

The total_charges column initially contained **11 blank records**, representing approximately **0.16% of the dataset**.

These records were associated with customers having zero tenure. Since Total Charges is a financial measure accumulated over the customer's relationship with the company, imputing these values using Monthly Charges would not reliably represent the actual recorded value. Therefore, the 11 records were removed from the analytical dataset.

This reduced the dataset from:

7,043 records → 7,032 records

Data Type Conversion

The following numerical variables were explicitly converted into numeric data types:

- tenure
- monthly_charges
- total_charges

This ensured that the variables could be correctly used for statistical calculations, correlation analysis, and numerical visualizations.

Column Standardization

The original column names were standardized using the **snake_case** naming convention.

For example:

TotalCharges → total_charges

MonthlyCharges → monthly_charges

This improved consistency and readability throughout the Python analysis.

Outlier Detection

The numerical variables were examined using the **Interquartile Range (IQR) method**.

The following variables were checked:

- Tenure
- Monthly Charges
- Total Charges

No significant outliers were identified in these numerical variables. Therefore, no observations were removed specifically because of outlier values.

Final Analytical Dataset

After completing the cleaning and preparation process, the final dataset contained:

7,032 customer records and 21 columns.

The cleaned dataset was then used for the subsequent univariate, numerical, and bivariate churn analyses.

6. Exploratory Data Analysis (EDA)

The exploratory analysis was performed to understand **which types of customers are more likely to churn and which customer characteristics are associated with greater retention**.

The analysis was divided into three major areas:

- **Univariate Churn Analysis**
- **Numerical Analysis**
- **Bivariate Churn Analysis**

Only the strongest and most business-relevant findings are highlighted below.

6.1 Overall Churn Distribution

Approximately **27% of customers have churned**, while the remaining **73% have remained with the company**.

This indicates that churn represents a significant customer-retention challenge, with approximately **one in every four customers** leaving the company.

Business implication:

The company has a substantial opportunity to improve retention by identifying high-risk customer segments and addressing the factors associated with churn.

6.2 Key Customer Characteristics Associated with Churn

1. Contract Type — Strong Churn Indicator

Customers on **month-to-month contracts** show substantially higher churn compared with customers on longer-term contracts.

Customers with **two-year contracts** demonstrate the lowest churn tendency.

Insight: Contract commitment appears to be strongly associated with customer retention.

2. Customer Tenure — Early-Stage Customers Are High Risk

Customers in the **0–12 month tenure group** have the highest churn rate at approximately **48%**.

In contrast, customers in the **61–72 month group** have a churn rate of approximately **7%**.

Churn decreases as customer tenure increases.

Insight: The early stage of the customer lifecycle represents the highest-risk period, while long-tenure customers are considerably more stable.

3. Internet Service — Fiber Optic Customers Show Higher Churn

Customers using **fiber optic internet** show higher churn compared with customers using DSL or no internet service.

The pattern becomes particularly important when fiber optic service is combined with other risk characteristics.

Insight: Fiber optic customers should be examined more closely for pricing, service quality, or customer-experience issues.

4. Technical Support & Value-Added Services

Customers without services such as:

- Online Security
- Online Backup
- Device Protection
- Tech Support

show higher churn tendencies.

The absence of **Tech Support** is particularly notable.

Insight: Customers who do not use additional support and protection services may have weaker engagement with the company's service ecosystem and may represent an important retention opportunity.

5. Payment Method — Electronic Check Customers Show Higher Churn

Customers using **electronic check** as their payment method demonstrate a higher churn tendency compared with customers using other payment methods.

Insight: Payment-method preference may be associated with customer behavior or other underlying characteristics that contribute to churn.

6. Customer Profile — Demographic and Household Characteristics

Several customer characteristics were associated with differences in churn:

- **Senior citizens** show higher churn than non-senior customers.
- Customers **without partners** show higher churn.

- Customers **without dependents** show higher churn.

Gender, however, shows little meaningful difference in churn.

Insight: Household structure and age-related characteristics appear more informative for churn analysis than gender.

6.3 Numerical Analysis

Tenure

The distribution of tenure shows a clear difference between churned and active customers.

Churn is concentrated among **short-tenure customers**, while customers who remain with the company for longer periods are considerably more stable.

Monthly Charges

Customers with **higher monthly charges** show greater churn tendency.

This suggests that pricing or perceived value may be important factors in the customer's decision to remain with the company.

Total Charges

Churned customers are concentrated toward the **lower end of Total Charges**.

This is largely connected to their shorter tenure: customers who leave early have had less time to accumulate total charges.

Therefore, Total Charges should not be interpreted independently as a direct cause of churn.

6.4 Combined Customer Risk Pattern

One of the most important findings from the analysis is that churn is not driven by a single characteristic.

A particularly high-risk customer profile emerges when several characteristics occur together:

Short tenure + high monthly charges + month-to-month contract + fiber optic internet + limited support/add-on services

These characteristics represent a customer segment that should receive greater attention from the retention team.

Key Takeaway

Customer churn is concentrated among early-stage, higher-paying, less contract-committed customers, particularly when they use fiber optic services and lack additional support or protection services.

The analysis therefore suggests that the company should prioritize **early customer engagement, contract conversion, service-value improvement, and targeted retention strategies** rather than applying the same retention approach to every customer.

7. Business Recommendations

The analysis identified several customer characteristics associated with higher churn. Based on these findings, the following actions are recommended to improve customer retention.

1. Strengthen Early-Stage Customer Retention

What: Prioritize customers during their first 12 months with targeted onboarding and engagement programs.

Why: Customers in the **0-12 month tenure group have the highest churn rate (~48%)**, while churn decreases substantially as tenure increases.

Recommendation: Introduce structured onboarding, early satisfaction checks, proactive support, and targeted retention offers during the first year of the customer relationship.

2. Encourage Customers to Move Beyond Month-to-Month Contracts

What: Encourage suitable month-to-month customers to transition to longer-term contracts.

Why: Month-to-month customers show substantially higher churn, while two-year contract customers demonstrate much stronger retention.

Recommendation: Offer appropriate incentives such as discounts, additional services, or loyalty benefits for customers willing to commit to longer-term contracts.

3. Review the Value Proposition for High-Charge Customers

What: Identify customers with relatively high monthly charges and evaluate their perceived value.

Why: Higher monthly charges are associated with higher churn, particularly among customers with short tenure.

Recommendation: Review pricing, plan benefits, and customer experience for high-charge customers and provide targeted plan reviews or personalized offers where appropriate.

4. Investigate Fiber Optic Customer Churn

What: Conduct a deeper review of customers using fiber optic internet.

Why: Fiber optic customers demonstrate higher churn than customers using other internet-service categories.

Recommendation: Investigate potential issues related to pricing, service quality, support experience, or perceived value and develop targeted retention strategies for this segment.

5. Promote Value-Added Support and Protection Services

What: Increase adoption of services such as Tech Support, Online Security, Online Backup, and Device Protection.

Why: Customers without these services show higher churn tendencies, with the absence of Tech Support being particularly notable.

Recommendation: Use onboarding and customer-engagement campaigns to communicate the benefits of these services and offer relevant bundles or targeted incentives.

6. Target High-Risk Customer Combinations

What: Develop retention strategies based on combinations of customer characteristics rather than relying on a single churn indicator.

Why: The analysis shows that churn risk becomes more concerning when characteristics such as **short tenure, high monthly charges, month-to-month contracts, fiber optic service, and limited support services** occur together.

Recommendation: Create a high-risk customer segment using these characteristics and prioritize proactive retention campaigns for customers within this group.

7. Examine Electronic Check Customers

What: Review customers using electronic check as their payment method.

Why: Electronic-check customers demonstrate a higher churn tendency than customers using other payment methods.

Recommendation: Investigate whether payment experience, customer profile, or other underlying factors are contributing to the observed relationship before implementing targeted interventions.

8. Conclusion

This Customer Churn Analysis used Exploratory Data Analysis to understand **why customers churn, who is more likely to churn, and which customer characteristics are associated with higher or lower churn risk.**

The analysis found that churn is particularly concentrated among **early-tenure customers, customers on month-to-month contracts, customers with higher monthly charges, fiber optic users, and customers without additional support or protection services.**

The strongest overall pattern is that **customer churn is influenced by a combination of lifecycle, pricing, contract, service, and customer-engagement characteristics rather than by one factor alone.**

The findings suggest that the company should focus particularly on **early-stage customers and high-risk customer combinations**, while strengthening long-term contract adoption and the perceived value of additional services.

These findings provide a practical foundation for developing targeted customer-retention strategies. However, because this project is an **EDA study**, the identified relationships should be treated as associations rather than proof of causation. Further analysis using statistical testing, customer-level churn prediction, and controlled retention experiments could help validate and quantify the impact of these factors.

Final Business Takeaway

The greatest retention opportunity lies in identifying high-risk customers early, understanding the combination of factors associated with their churn, and taking proactive action before they leave.